Stanley Nelavala

Software Engineer, AI Platform: Agents, MCP, Evals and Inference Infrastructure

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Huntington, IN | Open to relocate anywhere

Summary

Full-stack engineer who builds AI platforms: a public LangGraph and MCP agent platform with 62 deterministic replay evals and LLM failover, a Go inference service (ONNX, int8) sized for a 10x peak, and a Rust gateway. At a 10-person startup I own services across frontend, backend and AWS infrastructure. I use Claude Code, Codex and opencode every day and still read and debug every line. M.S. in Computer Science (4.0 GPA). AWS Certified Solutions Architect, Associate.

Experience

Software Development Engineer Nov 2025 – Present
Mass Technologies, team of 10 (RxApps360 hospital platform, pre-launch; numbers from dev and test) *Huntington, IN*

- Designed, built and integrated into every platform module the signed service-to-service API (M2M): each call proves which service and tenant sent it. After a security audit, hardened it with a Redis rate limiter and replay protection.
- Built RxDesign, the hospital design service (FastAPI, React, TypeScript), with a deterministic, rule-based layout engine and an LLM setting; 220+ offline API tests run with fake models, so tests never call an LLM.
- Automated weekday start and stop of dev and acceptance servers and databases (ECS, RDS): about 53 hours a week instead of 168.
- Replaced offset paging with keyset paging on the catalog device list: row 150,000 of a 328,000-device dev database fell from 285 ms to 0.20 ms, with no skipped or repeated rows (tested with inserts, deletes and ties).
- Built the RxAsset iOS and Android app (React Native, Expo) with scan-to-search and an on-device voice assistant (speech runs on the phone, no cloud call); set up release pipelines in GitHub Actions with automatic TestFlight submission.
- Standardized malware scanning of file uploads (GuardDuty) across 9 app modules.
- Use Claude Code, Codex and opencode every day to design, write, debug and review code; tests and written acceptance checks gate every change.

Adjunct Faculty (part-time) Jan 2026 – Aug 2026
The College of St. Scholastica *Duluth, MN*

- Taught Programming II (Java) and Machine Learning for Data Science.

Graduate Research Assistant and Teaching Assistant Aug 2023 – Aug 2025
University of Minnesota Duluth *Duluth, MN*

- Built a pipeline (Python, Node.js, MySQL, FFmpeg, PaddleOCR, BERTopic) that analyzed 1M+ TikTok videos from 300k+ authors for thesis research on misinformation.

AI projects (public code, tests)

Agent Integration Platform | *Python, FastAPI, LangGraph, MCP, PostgreSQL (pgvector), Redis* GitHub

- Multi-step agent with typed state, one constrained MCP write tool, human approval that resumes after restart, and step and token budgets; hybrid retrieval with access labels.
- 62 deterministic replay evals, prompt-injection tests, output guardrails, PII redaction, an append-only audit log; failover between two LLM adapters with a circuit breaker.

Inference Gateway | *Go, gRPC, ONNX Runtime, PyTorch, OpenTelemetry, Helm* GitHub

- Go service for an int8-quantized classifier with micro-batching, deadlines and load shedding; macro-F1 0.8636 for fp32 and int8; sized for a 10x peak at 2 to 3 replicas (local benchmark).

Webhook and Connector Gateway | *Rust, axum, tokio, PostgreSQL, SQLite, OpenTelemetry* GitHub

- Rust gateway: HMAC-signed webhooks, per-tenant limits, idempotency, durable store before delivery, retries, dead letters and circuit breakers; 88 tests on Postgres and SQLite.

Skills

AI platform: LangGraph, MCP servers and tools, agent evals, guardrails, sandboxed runs, LLM failover, ONNX Runtime, PyTorch, int8 quantization
AI-assisted coding: Claude Code and subagents, Codex, opencode, a DeepSeek harness; planner, manager and worker runs with acceptance checks
Languages: Python, Rust, Go, TypeScript, SQL, Java
Infrastructure: AWS (ECS, RDS, S3, SQS), Terraform, Docker, Helm on kind, GitHub Actions, OpenTelemetry, Prometheus
Full stack: FastAPI, React, React Native, PostgreSQL, Redis, gRPC

Education and Certification

M.S. Computer Science, University of Minnesota Duluth, 2023 – 2025, GPA 4.0, Graduate Summer Research Fellowship 2024 and 2025
B.Tech Computer Science and Engineering, IIIT D&M Kancheepuram, 2019 – 2023
AWS Certified Solutions Architect, Associate, Feb 2025 – Feb 2028 (verifiable on Credly)